Repairing a broken reasoning policy with on-policy distillation

Research idea #1 · OPD self-teacher recovery · 2026-05-29

Question

- A prior result: wrong-reasoning SFT degrades a model's reasoning POLICY globally (not just memorized poison — wrong-answer adoption stays <=5%).
- On-policy distillation (Thinking Machines, 2025) recovers post-trained behavior by sampling from the student and grading each token against a frozen teacher.
- Does the same self-teacher loop repair REASONING after a poison SFT breaks it?

TL;DR — yes, decisively

- OPD against the pre-damage teacher recovers every damage level to ~the teacher ceiling on held-out GSM8K-test, using NO labeled reasoning data and no reward model.
- It beats the obvious off-policy fix (continue clean SFT) on OOD in every condition (+7 to +23pp); clean SFT even DEGRADES OOD for the underwrite conditions.
- Recovery is capability, not memorization: wrong-answer adoption stays ~0.
- Works for subtle (~10pp) and catastrophic (~30pp OOD) damage alike.

Setup & method

The OPD repair recipe

- Student = base + poison-SFT adapter, fused, with a fresh trainable LoRA on top.
- Teacher = the original Qwen2.5-Math-1.5B-Instruct (frozen) — the 'pre-damage self'.
- Sample completions FROM the student on GSM8K-train prompts (on-policy).
- Score every token with one teacher forward pass; advantage = -(student_logprob - teacher_logprob).
- Policy-gradient update (single inner step = unbiased reverse-KL gradient).
- No reward model, no labels, one teacher forward pass per rollout.

Conditions (the damage to repair)

- underwrite_010: SFT on 10 wrong + 90 clean (subtle damage)
- underwrite_050: SFT on 50 wrong + 50 clean (moderate)
- overwrite: SFT on 100 wrong (catastrophic)

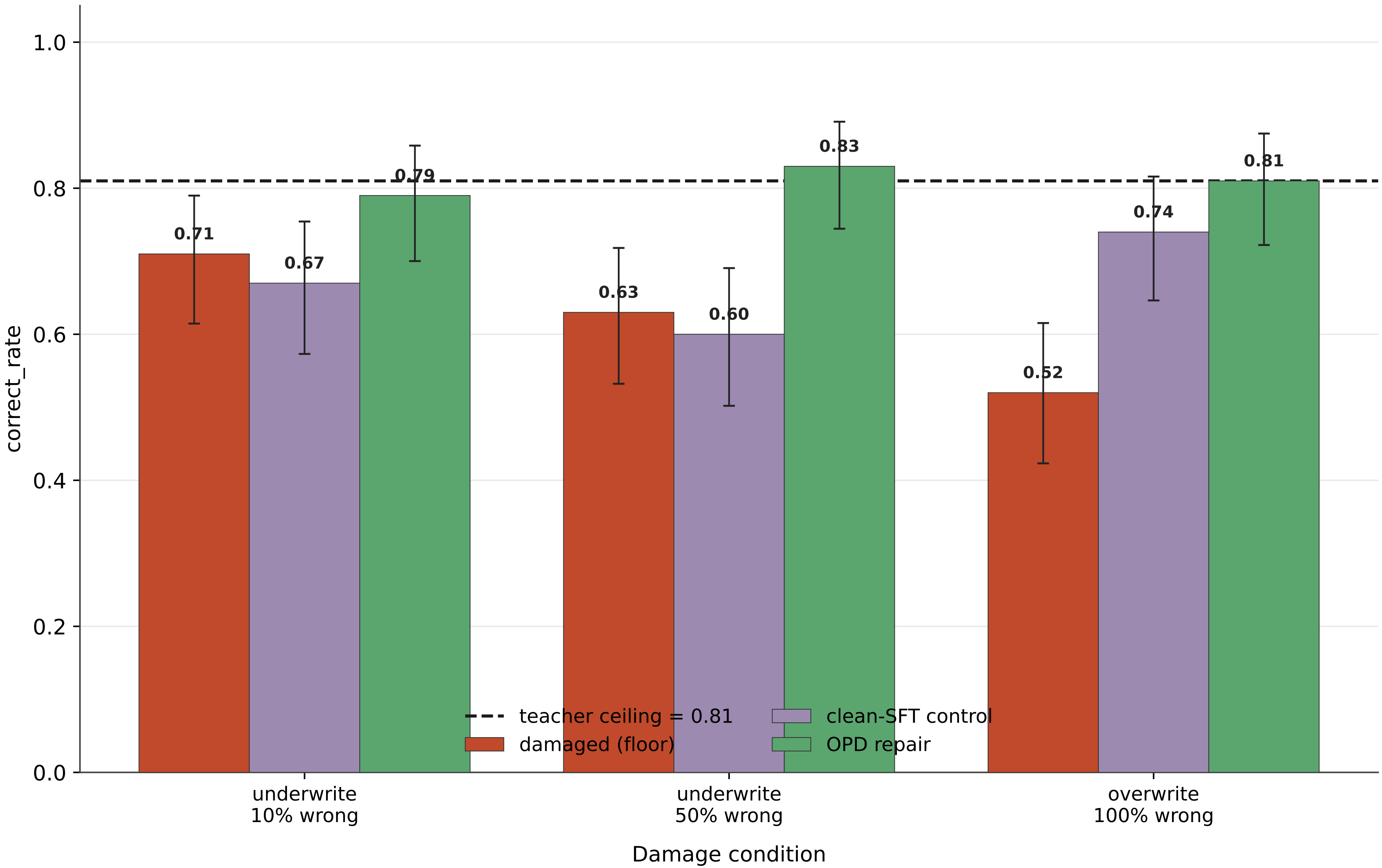
Baselines & metrics

- Teacher (ceiling) | damaged (floor) | OPD repair | clean-SFT control (H2 off-policy baseline).
- correct_rate (+95% Wilson CI) on in-distribution (calc-error) and OOD (GSM8K-test).
- wrong_adoption_rate: did the model reproduce the planted wrong answers? (capability vs memorization).
- OPD training prompts are GSM8K-TRAIN — disjoint from both eval sets (no contamination).

Clean-SFT control = continue LoRA SFT on clean reasoning data on top of the damaged model.

OOD recovery: GSM8K-test (the honest measure)

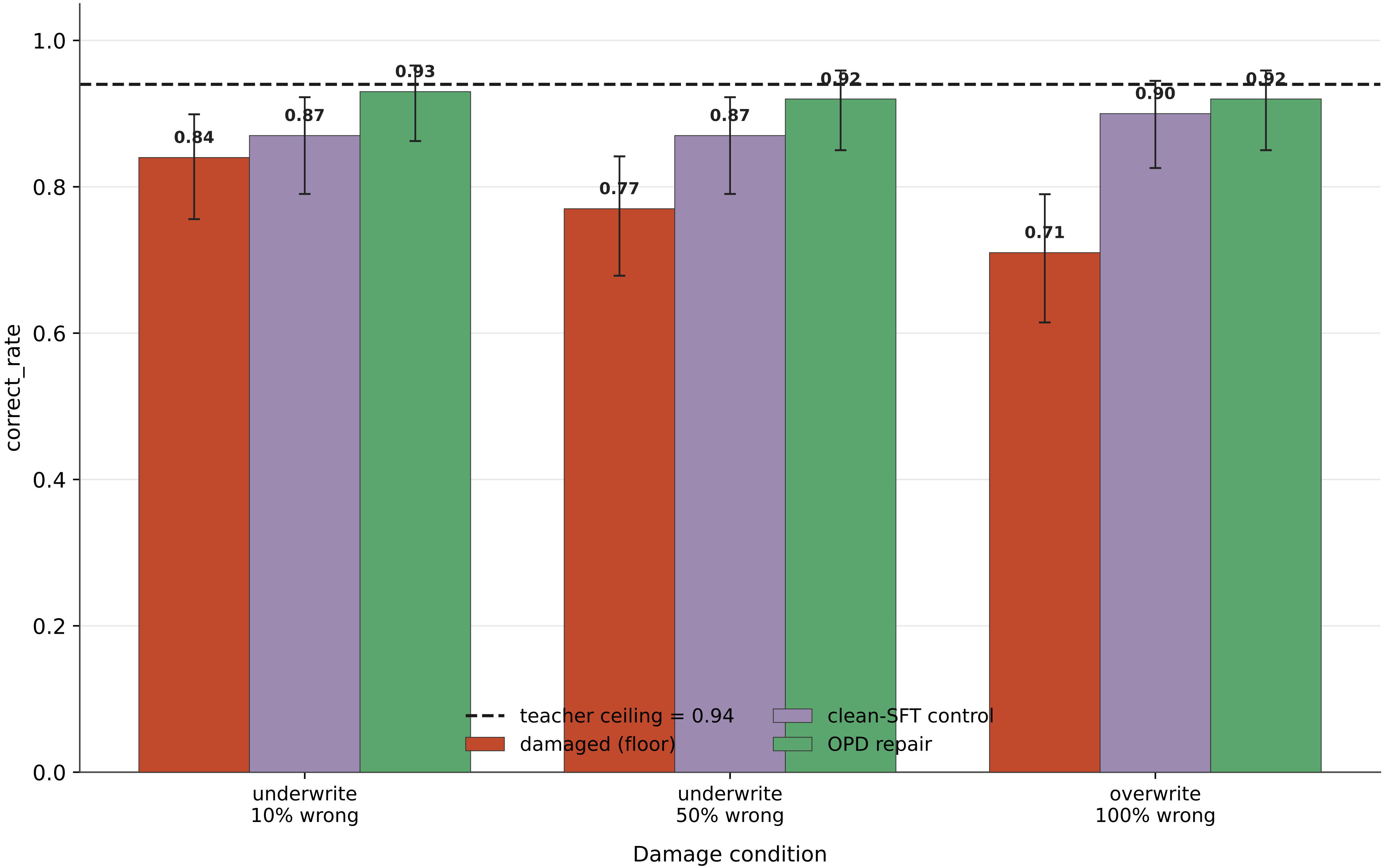
Held-out real benchmark · bars = correct_rate · whiskers = 95% Wilson CI



OPD recovers to the ceiling in every condition (overwrite 100% of the gap closed). Clean-SFT lags badly and even drops below the damaged floor for the underwrite cases.

In-distribution recovery (calculation-error eval)

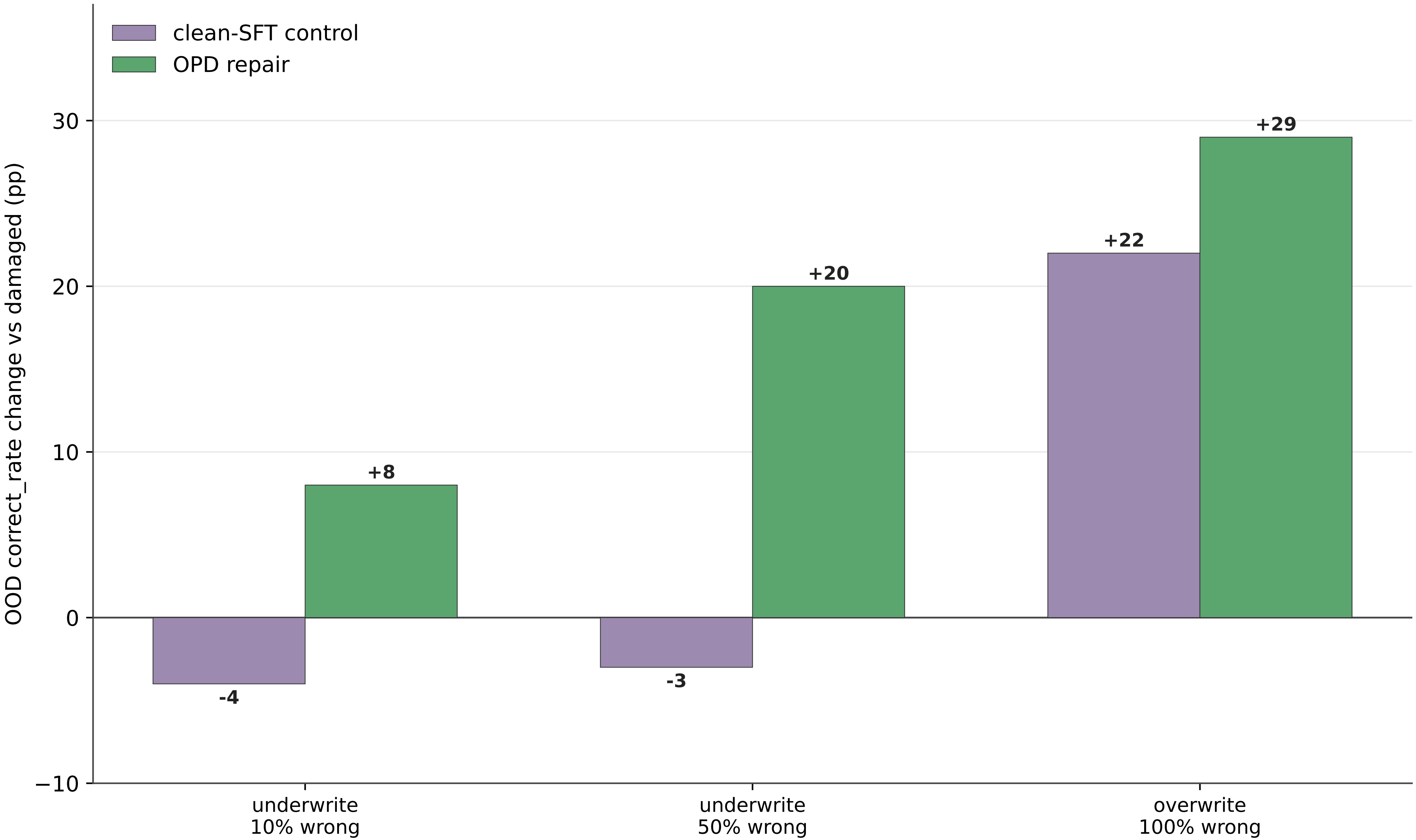
Same generator family as the SFT data · bars = correct_rate · whiskers = 95% Wilson CI



OPD wins here too, but clean-SFT looks deceptively good — this eval is stylistically close to the clean SFT data, which is exactly why OOD is the trustworthy number.

H2: OPD beats the off-policy baseline on OOD

Change in GSM8K-test correct_rate vs the damaged model (percentage points).



On held-out GSM8K-test, continuing CLEAN SFT (off-policy) barely helps and even HURTS for the underwrite conditions, while OPD (on-policy) recovers to the teacher ceiling. More off-policy SFT keeps the model off-policy; on-policy sampling repairs the policy.

Takeaways

What the data shows

- H1 — OPD repairs the reasoning policy. Full recovery to the teacher ceiling on OOD, across all three damage levels, with no labeled data and no reward model.
- H2 — on-policy beats off-policy. OPD > clean-SFT on OOD in every condition (+7 to +23pp). Continuing clean SFT even degraded OOD for underwrite_050 (0.63->0.60) and _010 (0.71->0.67).
- This is the compounding-error / distribution-shift story made concrete: more off-policy SFT keeps you off-policy; sampling from the student and correcting on-policy repairs the policy.
- Capability, not memorization: wrong-answer adoption ~0 for every repaired model.
- Severity-independent: subtle (~10pp) and catastrophic (~30pp) damage both recover fully.

Caveats

- 1 seed; Wilson CIs are ~+/-9pp wide. Multi-seed needed to tighten claims.
- Single model / single task family (GSM-style math, Qwen2.5-Math-1.5B).
- Teacher == exact pre-damage model (best case). Weaker/standalone teachers untested.

Pipeline: opd_repair_experiment/ (run_experiments.sh). All artifacts gitignored.

Proposed follow-up experiments

Tighten the core claim

- E1. Multi-seed (3-5) sweep over all conditions — turns the bars into significance.
- E2. OPD dose-response: steps (10/25/50/100/200) and prompt-pool size (1/8/64/256). TM claim that even ~1 prompt works — test the data-efficiency floor here.

Probe the mechanism / limits

- E3. Weak-teacher OPD: teacher = a clean-SFT model (not the pristine base). Tests the 'student can only be as good as the teacher' ceiling.
- E4. Self-repair: teacher = an EMA / earlier checkpoint of the damaged student itself. Does the model bootstrap recovery with no external pre-damage checkpoint?
- E5. Forward-KL vs reverse-KL repair: does mode-covering forward KL recover broader coverage (fewer rare-case failures) at the cost of sharpness?

Push toward a stronger result

- E6. Hybrid objective: OPD per-token KL + sparse correctness reward (research idea on combining distillation + environment reward). Does it exceed the teacher ceiling?
- E7. Cross-task transfer: repair on GSM-train prompts, eval on MultiArith / GSM-Hard — does OPD restore broad reasoning or just GSM8K-shaped problems?
- E8. Repair a DIFFERENT damage type (instruction-following / format poison) to show the loop is damage-agnostic, not GSM-specific.
- E9. Scale check: same protocol on Qwen2.5-Math-7B to confirm it isn't a small-model artifact.

Highest value next: E1 (seeds) for rigor, then E3 (weak teacher) for the most novel insight.